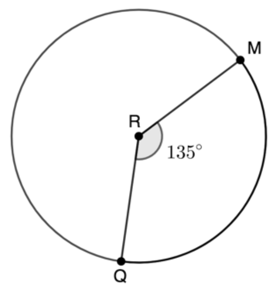
Geometry
Unit 10
Lesson 2 Homework

Name___Answer Key_______
Date____________________
Period__________________

For questions 1-2, Circle $R$ is shown where points $M$ and $Q$ are on the circle.




1. What is the ratio of the central angle, $\angle QRM$, to the whole circle? Simplify the ratio.

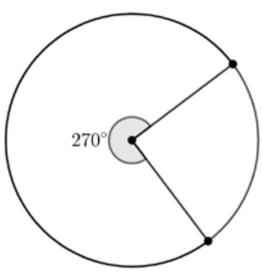
$\frac{135}{360} = \frac{3}{8}$

2. What fraction of the circle's circumference does $\widehat{MQ}$ represent?

$\frac{225}{360} = \frac{5}{8}$ for the larger arc; $\frac{135}{360} = \frac{3}{8}$ for the smaller arc

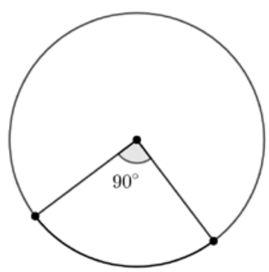
For questions 3-8, find the ratio of the arc length to the circumference of the circle.

3.



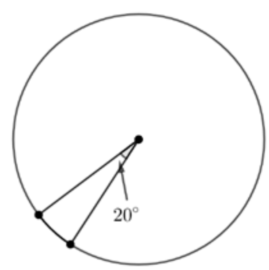

$\frac{270}{360} = \frac{3}{4}$

4.



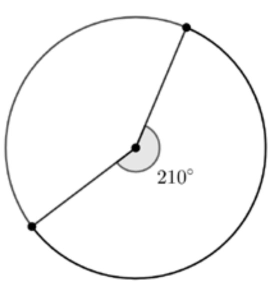

$\frac{90}{360} = \frac{1}{4}$

5.



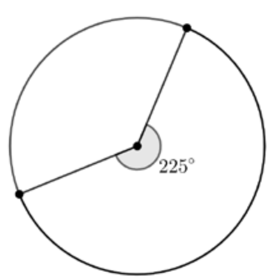

$\frac{20}{360} = \frac{1}{18}$

6.



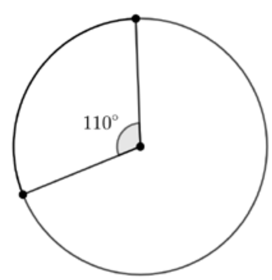

$\frac{210}{360} = \frac{7}{12}$

7.




$\frac{225}{360} = \frac{5}{8}$

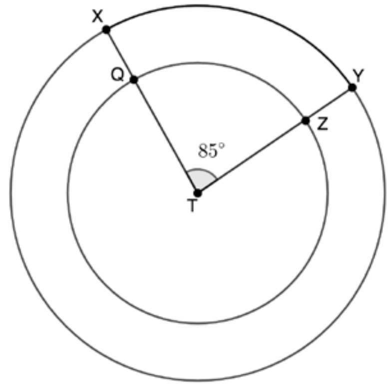
8.




$\frac{110}{360} = \frac{11}{36}$

For questions 9-10, consider the following information.

Points $X$ and $Y$ lie on Circle $T$. The radius $\overline{TY}$ measures $8$ inches (in.). A smaller Circle $T$ with a radius measure of $5$ in. rests inside the larger Circle $T$, where points $Q$ and $Z$ lie on the smaller circle. The smaller Circle $T$ has central angle $\angle QTZ$ that measures $85°$ and the larger Circle $T$ has central angle $\angle XTY$ that measures $85°$.




9. What is the ratio of the arc length to circumference of Circle $T$ with radius measure of $8$ in.?

$\frac{85}{360} = \frac{17}{72}$

10. What is the ratio of the arc length to circumference of Circle $T$ with radius measure of $5$ in.?

$\frac{85}{360} = \frac{17}{72}$